



INSTRUCTION DIVISION

FIRST SEMESTER 2026-27

Course Handout Part II

Date: 3/08/2026

In addition to Part-I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : MPBA G515
Course Title : **PRESCRIPTIVE ANALYTICS WITH MATHEMATICAL PROGRAMMING**
Instructor-in-charge : **PROF RAJESH MATAI**

1. Scope & Objective:

Prescriptive analytics is the third and final phase of business analytics, which also includes descriptive and predictive analytics. Referred to as the "final frontier of analytic capabilities", prescriptive analytics entails the application of mathematical and computational sciences and suggests decision options for how to take advantage of the results of descriptive and predictive phases.

The first stage of business analytics is descriptive analytics, which still accounts for the majority of all business analytics today. Descriptive analytics looks at past performance and understands that performance by mining historical data to look for the reasons behind past success or failure. Most management reporting – such as sales, marketing, operations, and finance – uses this type of post-mortem analysis.

The next phase is predictive analytics. Predictive analytics answers the question of what is likely to happen in the future. This is where historical data is combined with rules, algorithms, and occasionally external data to determine the probable future outcome of an event or the likelihood that a situation will occur.

The final phase is prescriptive analytics, which goes beyond predicting future outcomes to suggest actions to benefit from those predictions and show the implications of each decision option.

While the term prescriptive analytics was first coined by [IBM](#) and was later [trademarked](#) by Texas-based company Ayata, the underlying concepts have been around for hundreds of years in the form of Operations Research. Prescriptive Analytics has been considered the next step towards increasing data analytics maturity and leading to optimized, ahead-of-time decision-making for business performance improvement. New Information and Communication Technologies (ICT), such as the Internet of Things (IoT), real-time streaming, and sensor-driven business operations, have bolstered prescriptive analytics and provided businesses with accurate prescriptions, facilitating reliable and effective decision-making.



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Prescriptive analytics assists the decision-maker in identifying the best action (optimal solution) given the problem context. The difference between decisions made using descriptive/predictive analytics and prescriptive analytics is that a prescriptive analytics algorithm seeks the best decision (optimal solution) based on an objective function (sometimes multiple objective functions) and a set of constraints. Prescriptive analytics models attempt to solve complex optimization problems of the modern era.

Students will be exposed to a powerful optimization software package, Lingo, so they can formulate and solve real-life Prescriptive Analytics problems. Students can download a trial version (free) of Lingo from <https://www.lindo.com/index.php/products/lingo-and-optimization-modeling>. Using Lingo, large-scale, complex Prescriptive Analytics problems can be solved.

2. Course Learning Outcomes:

- Students will be able to formulate prescriptive analytics real-life problems and use their applications.
- Students will learn the basics of Linear Programming, Transportation and Assignment problems, Integer Programming, Waiting Line Problems, Simulation modeling, and Goal Programming
- Students will be able to formulate real-life large-sized problems in Lingo software and solve them. They will also learn how to interpret Lingo results and apply them to real-world decision-making for prescriptive analytics problems.

3. Textbooks and References:

- TB1. Taylor, Bernard W.; Introduction to Management Science, Pearson Education, India. 9th Edition or above.
- TB2. Render, B., Stair, Jr., R.M., Hanna, M.E., and Badri, T.N.; Quantitative Analysis for Management, Pearson Education, India, 10th Edition or above.
- RB1. Frederick S. Hillier and Mark S. Hillier; Introduction to Management Science: modelling and case studies approach with spreadsheets, McGraw-Hill Irwin, 5th Edition or above.
- RB2. Taha, H. A. (2003). Operations research: an introduction. Pearson Education.

Some Reference links:

- Basu, Atanu (2019). "Five pillars of prescriptive analytics success". <https://pubsonline.informs.org/doi/10.1287/LYTX.2013.02.07/full/>
- "Gartner terms Prescriptive Analytics as the 'Final Frontier' of Analytic Capabilities | Globys.com".

- 4. Method of Conduct of Course:** The course will be taught through lectures, case studies, real-life applications, students' project presentations, and usage of lingo optimization software.



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5. Course Plan:

Module No. (Topic)	Lecture Session	Reference	Learning outcomes
(I) (i) Operations Research Approach to Problem Solving	1-2	Chapter 1 (T1)	The Operations Research Approach
Linear Programming (LP): Model Formulation and Graphical Solution	3-6	Chapter 2 (T1)	Model Formulation, A Maximization Model, Graphical Solutions of Linear Programming Models, A Minimization Model, Irregular Types of Linear Programming Problems, Characteristics of Linear Programming Problems
(ii) Linear Programming: Computer Solution and Sensitivity Analysis	7-9	Chapter 3 (T1)	Computer/ Software Solution, Sensitivity Analysis
Solving Linear Programming (OR) Problems in Lingo	10-12	LINGO and optimization modeling (lindo.com)	Students will be able to formulate LP and OR problems and solve them in Lingo. Also, they will learn how to interpret software output for real-life decision-making.
(iii) Linear Programming: Modelling and Applications	13-16	Chapter 4 (T1) Chapter 8 (T2)	A Product Mix Problem, A Diet Problem, An Investment Problem, A Marketing Problem, A Transportation Problem, A Blend Problem, A Multiperiod Scheduling Problem, A Data Envelopment Analysis Problem, A Production Scheduling Problem, An Employees Scheduling Problem, Other Linear Programming Applications
(iv) Linear Programming: The Simplex Method	17-21	Chapter 9 (T2)	Convert LP constraints to equalities with slack, surplus, and artificial variables. Set up and solve LP problems with simplex tableaus. Interpret the meaning of every number in a simplex tableau. Recognize special cases such as infeasibility, unboundedness, and degeneracy. Use the simplex tables to conduct sensitivity





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			analysis.
(II) (i) Transportation and Assignment Problem	22-26	Chapter 10 (T2)	Structure special LP problems using the transportation and assignment models. Use the northwest corner, VAM, and MODI methods for solving the transportation problem. Solve assignment problems with the Hungarian (matrix reduction) method.
(ii) Integer Programming (IP) and Goal Programming (GP)	27-33	Chapter 11 (T2)	Understand the difference between LP and Integer Programming, Using Software to solve Integer Programming Problems (IPP), and Some IP Applications. Apply the branch and bound method to solve IPP. Solving Goal Programming Problems (GPP) graphically, Goal Programming with Weighted Goals, Using Software for solving GPP
(III) (i) Waiting Lines and Queuing Theory Models	34-37	Chapter 14 (T2)	Waiting Line Costs, Characteristics of a Queuing System, Single-Channel Queuing Model with Poisson Arrivals and Exponential Service Times (M/M/1), Multichannel Queuing Model with Poisson Arrivals and Exponential Service Times (M/M/m), Constant Service Time Model (M/D/1), Finite Population Model (M/M/1 with Finite Source), Some General Operating Characteristic Relationships
(ii) Simulation Modelling	38-40	Chapter 15 (T2)	Introduction, Advantages, and Disadvantages of Simulation, Monte Carlo Simulation, Simulation of a Queuing Problem, Role of Computers in Simulation
Some Advanced Applications of Prescriptive Analytics	41-42	Case studies/Research Papers	Students will be able to understand real-life applications of Prescriptive Analytics.



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6. Evaluation Components:

S. No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Remarks
1	Mid-Semester Examination	90 minutes	20% (40 Marks)	10/10/2026 (AN2)	Close Book
2	Mini Case Discussions (6 cases of 3% weight each)	During regular class hours	18% (36 Marks)	To be announced (Students have to participate in these mini case discussions. Anyone absent from class will get no credit in that case. Cases will be assigned time to time)	Open Book
3	Quiz (One)	25 minutes	7% (14 Marks)	Announced in class	Close Book
4	Mini Project		20% (40 Marks)	Students will work in groups on any Prescriptive Analytics Problem. Formulate and solve it in Lingo/suitable software, present the findings in a presentation, and submit a report at semester's end. Students can use databases available in the Library to solve a problem in the Marketing, Finance, Operations, Logistics, Supply Chain, or Health Care domains, or in any domain of their interest.	
5	Comprehensive Examination	180 minutes	35% (70 Marks)	16/12/2026 (AN)	Open Book

- Mini Case discussions:** Mini Cases will be assigned from time to time. Students must read the assigned case, conduct proper analysis, and come fully prepared for class discussions. It is highly desired that students attend all classes and contribute to case discussions. Any student absent from that class will not receive credit for that case discussion.
- Mini Project:** Students will take any real Prescriptive Analytics problem and do projects in groups. At the end of the semester, all groups will give a presentation and submit a report.
- Chamber Consultation Hour:** Friday after class.
- Notices:** All notices of this course will be displayed on the Department of Management Notice Board or Online.
- Make-up Policy:** Make-ups may be allowed only in genuine medical cases with prior permission of I/C. For Quiz and Mini Case discussions, no make-up will be given.
- Team Formation:** Students have to give mini-project presentations in a team. Teams will be formed by the Instructor.

Instructor-In-Charge



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